Cofinal witnesses and centered carriers for Diophantine consistency

Research prepared for Vladimir Reshetnikov

25 October 2026

Abstract

We construct explicit polynomial equations whose unsolvability in their stated integer domains is provably equivalent in ZFC to syntactic consistency of ZFC. The direct constructions combine common remainder certificates with an odd-square radix and exact affine elimination. A separate natural-domain construction uses an even descending-factorial horizon and a cofinal beta base, permitting a smaller witness system. For ten positive variables, a small-base Pell chart delays full denominator recovery until an exponential value has been established. Centered root carriers shorten two unrestricted-integer evaluators without changing variable count or degree. The complete equations, fixed numerical tuples, reconstruction maps, ordinary proofs and exact finite checks accompany this report. All comparisons retain their domain and all measured costs. Inherited calibrations are explicitly identified; no consistency decision, global minimality theorem or new end-to-end kernel certificate is claimed.

Contents

1 The objective and the comparison boundary **2**
1.1 Selected explicit equations . . . 3

2 Common remainders, odd-square radices, and 246-variable quartics **3**
2.1 The arithmetic composition: 252 to 250 . . . 4
2.2 Universal affine exposure: 250 to 246 or 247 . . . 4
2.3 A degree-three obstruction, not a general lower bound . . . 5
2.4 Reproducibility and the proof boundary . . . 5

3 A cofinal descending construction over the natural numbers **5**
3.1 Compatible positive centers and pooled low digits . . . 6
3.2 A new even-horizon cofinal beta specialization . . . 7
3.3 Exact compilation, verification, and selected tradeoffs . . . 9

4 Ten-positive equations: projection before full denominator recovery **9**
4.1 Complete polynomial definitions . . . 10
4.2 Roots, folding, and unconditional bounds . . . 11
4.3 Unshifted transport, including both signed exclusions . . . 11
4.4 Authenticate the first index before the full denominator . . . 12
4.5 Projection first, full denominator second . . . 12
4.6 Native soundness and constructive positive completeness . . . 13
4.7 Exact global degrees and reproducibility . . . 14

5 Complete fixed-coefficient recipe **15**

6 Centered integer-root charts: fewer operations at fixed degree **17**
6.1 The complete arithmetic interface . . . 17
6.2 The fourteen-variable equation and all-zero maps . . . 18
6.3 The thirteen-variable centered norm . . . 18
6.4 The exact shear and actual degree witnesses . . . 20
6.5 A scoped obstruction to degree reduction by this method . . . 20
6.6 Verification and interpretation . . . 20

7 Reproduction and the exact proof boundary **21**

A A separately scoped whole radix47 specialization **21**
A.1 The exact incoming source and its changed native theorem . . . 22
A.2 An explicit positive bridge, not a port-box substitution . . . 23
A.3 Constructive converse and exact finite closure . . . 24

B Complete complemented-label radix47 constants **24**

1 The objective and the comparison boundary

For each fixed polynomial $P \in \mathbb{Z}[x_1, \ldots, x_n]$ and stated domain D, our target is

$$\mathrm{ZFC} \vdash \left(\neg\exists \boldsymbol{x} \in D^n \; P(\boldsymbol{x}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

Here Con denotes the absence of a finite syntactic refutation in the specified effective proof system. We prove the equivalent solvability statement, rather than proving consistency or exhibiting an actual zero. The domain is part of the theorem: unrestricted, nonnegative and strictly positive integers cannot be interchanged in a complexity comparison.

The immutable comparison snapshot is `3ecd6a6cf53d47519228328567f45fb7ee9781d9` of ProveIt. This includes the seven parallel incoming packages and our preceding *Residual Bases and Denominator Engineering* package. The new 252-variable common-remainder system and cheaper compact circuits are used as comparators instead of advertising obsolete improvements over our previous report alone. Incoming placement is distinguished from repository adoption and from independent mathematical verification. The supplied baseline review authenticates the actual archive contents, full direct polynomial expansions, and coherent fixed tuples.

We report variable counts, total degree, residual count where applicable, collected support, maximum absolute coefficient, and live binary arithmetic operations. Each addition, subtraction and multiplication costs one, including a square and multiplication by a fixed constant; reused registers are free. A short evaluator need not have small expanded coefficients. Fixed ports are not quantified variables, but all their numerals and their coherent binding remain part of the specification.

1.1 Selected explicit equations

Domain and profile	Variables	Degree	Rows	Terms	Height
$\mathbb{Z}$, dimension	246	4	178	12541	12227
$\mathbb{Z}$, sparse	247	4	179	4352	23316
$\mathbb{Z}$, balanced	247	4	179	4432	17235
$\mathbb{Z}$, lower height	247	4	179	5648	6144
$\mathbb{N}$, cofinal dimension	219	4	171	8267	130686428
$\mathbb{N}$, cofinal sparse	221	4	173	5253	65867916
$\mathbb{N}$, noncofinal balanced	221	4	172	5586	65892620

The incoming common-remainder integer comparison has 252 variables, 182 rows, 5340 terms and height 128446. The new 247-variable sparse and balanced points improve all four varying measures; the 246-variable point reduces dimension further at a support cost. The low-height point is a different support/height tradeoff. In the natural branch, the new 219-variable point improves dimension and support against the incoming 221-variable/172-row/9536-term/height-83954186 construction, but has larger height. The noncofinal 221-variable form improves support and height at unchanged dimension and residual count. These are not claims of global variable-count records.

Domain and construction	Variables	Exact degree	Binary operations
Positive, partially normalized	10	382	79
Positive, smaller base	10	338	80
Unrestricted, centered two-root carrier	14	88	110
Unrestricted, centered three-root carrier	13	156	117

The positive 80/338 profile improves the matched incoming 80/382 operation/degree point in degree; the 79/382 profile saves one operation at that degree. The previous 14-integer degree-86/112-operation point remains a separate tradeoff. The new degree-88/110-operation point improves the former degree-94/110-operation choice, while the 13-integer degree-156 evaluator saves eight operations from 125 after an explicit integer coordinate shear; this is not an identical-polynomial recompilation in the original coordinates. The positive component retains two explicitly dominated control charts. All small-variable tuples include their complete fixed coefficients; expanded support and expanded coefficient height are not measured for these DAG representations.

2 Common remainders, odd-square radices, and 246-variable quartics

The strongest incoming literal integer source has 252 variables and 182 quadratic residuals. We compose its common-remainder normalization with the separately established odd-square radix construction, then transport the previous affine-exposure certificates and optimize the residual basis. This is a composition of identified arithmetic and universal algebraic proofs, not a change from unrestricted integers to natural variables. The completed profiles are:

Profile	Variables	Rows	Monomials	Height
Fewest variables	246	178	12,541	12,227
Sparse	247	179	4,352	23,316
Balanced	247	179	4,432	17,235
Low height	247	179	5,648	6,144

Each fixed equation is

$$P(\mathbf{x}) = \sum_{i=0}^{m-1} R_i(\mathbf{x})^2 = 0, \qquad \mathbf{x} \in \mathbb{Z}^n,$$

with every R_i quadratic or affine and P of exact degree four. The artifact supplies every residual, every collected coefficient, and the complete coordinate map. Height is the maximum absolute *collected* coefficient of P, not the height of its residuals. These are achieved tradeoffs; no global minimality is claimed.

2.1 The arithmetic composition: 252 to 250

Let q be the actual radix and B the exact-binomial output. In the centered incoming coordinates $q = x_{107} - 1$ and $B = x_5$. The only changed residual is the old ternary guard

$$-2 - x_5 + x_{107} - x_{243}^2 - x_{244}^2 - x_{245} - x_{245}^2 = 0.$$

Its three auxiliary coordinates occur nowhere else. Delete them and introduce a single unrestricted z with

$$x_{107} - 2 - 4z - 4z^2 = 0, \quad \text{equivalently} \quad q = (2z+1)^2.$$

All other 181 rows are unchanged after exact renumbering. This gives a 250-variable, 182-row source. The finite edit is not a pointwise bijection: the soundness and completeness arguments are semantic and are supplied separately in full.

Here is the essential noncircular soundness order. The unchanged signed frame, anchor, and pooled orientation first recover the canonical Pell pairs. The known positive horizon interprets $V = (F^2+1)^t$, including the base-one case. The two retained ceiling guards then give $F \geq 1$, $B \geq 2$, $V < B^2$, and $y = V(V-1) > F+2$. The latter is an actual source equation, not merely a cofinal choice. The new square gives $q \geq 1$, so numerator requests at base $q+1 \geq 2$ force every actual CRT representative nonnegative, whence $H \geq 1$. The existing request at base $BH \geq 2$ yields $x \geq 0$ and $q = (BH)^x$. The endpoint equations force $x \geq F+1 \geq 2$ and $x \geq 2 + (t+1)y$. Thus $q \geq B^2 > B$, recovering the removed output-slack inequality before binomial recovery. The normalized-remainder proof now applies unchanged.

For completeness, the cofinal beta moduli are odd because $y = V(V-1)$ is even; an even beta code x can therefore be chosen by CRT. The existing odd-modulus representative freedom permits H odd while preserving the opposite parities needed by the semantic anchor and all dominance bounds. Then BH is odd, x is even, and $z = ((BH)^{x/2} - 1)/2$ is integral. The genuine remainder bounds $r_i < L_i/32$ are unaffected. Their normalized representatives satisfy

$$4\sum_i (Mr_i/L_i)^2 < \frac{11}{256} M^2 < M^2,$$

so the single common norm guard still has its ternary witnesses. These two directions compose with the inherited C222/ZF/ZFC-refutation calibration. Hence ZFC proves that integer unsolvability of every final fixed quartic is equivalent to Con(ZFC). No consistency assertion or new end-to-end proof-assistant verification is being made.

2.2 Universal affine exposure: 250 to 246 or 247

For residuals, $R_i \leftarrow R_i + kR_j$ with $i \neq j$ is inverted by $-k$ and preserves the common zero set. Integral triangular coordinate shears are lattice bijections. A unit affine equation $\epsilon x_t + L = 0$,

$\epsilon \in \{-1, 1\}$, permits deletion and unique restoration $x_t = -\epsilon L$. These elementary facts preserve degree at most two and imply the same zero-set equivalence for the sum of squares. They do not imply equality of the old and new quartic polynomials.

After the six credited initial shears, exact expansion in the stronger 250-coordinate source gives

$$\begin{aligned} Q_{21} - Q_6 &= 4 + 4x_{10} + 8x_{11} + 4x_{21} - x_{22} + x_{37}, \\ Q_{73} - Q_{71} - 4Q_{70} - 8Q_{68} &= 12 - 6x_3 - 8x_{84} - 4x_{85} - x_{86} + x_{89}, \\ Q_{119} - Q_{117} &= 3 - 2x_{107} + 2x_{148} - x_{164} + x_{166}. \end{aligned}$$

These remove x_{37}, x_{89}, x_{166} and give 247 variables and 179 rows. The already affine row Q_{90} additionally removes x_{105}, giving 246 variables and 178 rows. Its dense substitution explains the larger support of the fewest-variable profile. A separately credited 76-step coordinate preconditioner is used for the other profiles; their current affine equations are verified exactly, not assumed to match these displays. Additional finite row and coordinate operations are frozen in each certificate. Every original source coordinate is printed as an affine polynomial in the final coordinates, and every original residual pullback is checked as a universal polynomial identity.

2.3 A degree-three obstruction, not a general lower bound

For the 178 residuals in 246 variables, form the integer matrix of the degree-two and degree-three coefficients of

$$R_i,\ x_0R_i, \ldots, x_{245}R_i \qquad (0 \le i < 178).$$

There are $178(246 + 1) = 43,966$ columns. An independent sparse elimination finds full column rank modulo the prime $1,000,003$. Therefore an integer minor is nonzero, and the matrix has full column rank over $\mathbb{Q}$. It follows that

$$\deg\left(\sum_i L_i R_i\right) \le 1, \quad L_i \text{ affine over } \mathbb{Q} \quad \Longrightarrow \quad L_i = 0 \text{ for all } i.$$

This rules out *all affine-polynomial multipliers*, including the nonlinear combinations x_jR_i, not only constant row combinations as in the prior round. It does not rule out higher-degree multipliers, semantic witness replacements, a different source circuit, or smaller quartics by other methods. A deliberately duplicated residual must lose rank.

2.4 Reproducibility and the proof boundary

The portable verification entry point rebuilds the inherited source, authenticates its retained files, checks the odd-square edit independently of its producer, verifies all source-restoration identities and quartic coefficients, and demands byte-identical fresh endpoint builds. Both ordinary Python and Python with optimization enabled are run. The residual text and TeX are parsed back to their exact coefficient lists; corrupt affine maps, coordinate shears, row operations, residuals, and expanded coefficients are rejected. The Macaulay computation uses exact finite-field arithmetic only. These checks establish the finite objects and universal algebra; the separately identified unbounded arithmetic and logical-calibration proofs remain their stated dependencies.

3 A cofinal descending construction over the natural numbers

All coordinates in this section range independently over $\mathbb{N} = \{0, 1, 2, \ldots\}$. The three selected equations are literal sums of squares of quadratic integer residuals. Their complete residual lists,

coordinate orders, expanded coefficient lists, source changes and restoration certificates accompany this report.

Endpoint	Variables	Rows	Quartic terms	Height
Round 18 balanced control	229	179	5,607	65,892,965
Incoming positive-center control	221	172	9,536	83,954,186
New `natural219`	219	171	8,267	130,686,428
New `sparse221`	221	173	5,253	65,867,916
New `balanced221`	221	172	5,586	65,892,620

The 219-variable equation is the new dimension endpoint. The 221-variable cofinal equation trades two coordinates for smaller support and height; the noncofinal 221-variable companion uses one fewer residual. These are measured constructions, not claims of global optimality.

Exact statement and proof boundary. For each selected residual list $R_0, \dots, R_{s-1}$, define the fixed ordinary polynomial

$$P(\mathbf{z}) = \sum_{j=0}^{s-1} R_j(\mathbf{z})^2.$$

Relative to the explicitly retained ordinary C222/ZF/ZFC calibration, P has a natural zero exactly when a finite syntactic ZFC refutation exists. Consequently ZFC proves that its no-natural-zero sentence is equivalent to Con(ZFC). This is not a proof of consistency or a newly kernel-checked end-to-end endpoint. The finite checkers establish the literal polynomials and exact algebra; the ordinary unbounded arguments establish their arithmetic interpretation. Factorials, binomials, variable powers and Pell sequences below occur only in that interpretation, never as primitives of the released equations.

3.1 Compatible positive centers and pooled low digits

The retained shift-free descending source has natural coordinates satisfying

$$n + 1 = F^2, \qquad V = BF + r, \qquad B = r + s + 1. \tag{N1}$$

The exact-power modulus row $T_n + n^2 + 1 = 2An$ excludes $n = 0$ before interpreting any power. Thus $F \geq 2$ and $n \geq 3$. The common frame and semantic Pell anchor authenticate six powers,

$$V = n^t, \quad N = B^x, \quad q = (NS)^x, \quad D_t = q^t, \quad D_y = q^{y+1}, \quad D_x = q^x, \tag{N2}$$

where S is one plus the nine positive independent CRT representatives. There are thirteen nonanchor seeds and eleven modular digit requests. The anchor index $H = J$ is the actual derived syntax-label sum.

First compose the incoming positive deterministic comparison-center construction with this descending source. Write its principal pair (X, Y), auxiliary pair (U, V_c), and comparison pair (S_c, R_c). Keep all three Pell norms and impose

$$V_c = 2Y^2(e + 1), \qquad T = U^2, \qquad C_c = A + (4Y + 1 - A)T, \tag{N3}$$

with natural $C_c - 1$, together with $S_c = X + Ug$ and $R_c = H + 4Yh$. The unchanged merged cap row gives

$$Y = H + \sum_i (Y_i - k_i)^2 + (A - 1)Q \geq H \geq 2$$

using only the raw frame $A \geq 2$ and the literal positive label expression. It precedes index interpretation. The auxiliary norm yields

$$T - 1 = 4(A^2 - 1)Y^4(e + 1)^2, \qquad T > A.$$

If $L = 4Y + 1 - A < 0$, then $C_c = A + LT < 0$, impossible. Hence $L \geq 0$, $C_c \geq A \geq 2$, and

$$C_c - 1 = 4YT - (A - 1)(T - 1) \equiv 0 \pmod{4Y}.$$

The old natural center witnesses are restored by

$$d = \frac{C_c - 1}{4Y} - 1, \qquad f = LU, \qquad e_{\text{old}} = 2e + 1.$$

Their nonnegativity follows on the retained equations; it is not claimed coefficientwise on the entire free orthant. Conversely, for canonical $Y = Y_H(A)$, choose a positive auxiliary Pell period modulo $2Y^2$. Then V_c is a positive multiple of $2Y^2$; taking comparison pair $(X_H(C_c), Y_H(C_c))$ supplies the remaining congruences and natural quotients. This uses fresh auxiliary witnesses, not a pointwise bijection on all old auxiliary tuples.

Next pool each equal-denominator family into one natural linear budget:

$$D = \sum_{i \in G} r_i + s_G + 1. \tag{N4}$$

The four groups have denominators $q^t, q^{t+1}, q^{y+1}, q^x$, and sizes $2, 5, 2, 2$. Soundness restores each old slack by the nonnegative polynomial $s_i = s_G + \sum_{j \neq i} r_j$, without any power or binomial semantics. The numerator rows that orient the consecutive Pell seeds remain unchanged.

The separate factorial-output bound $q = B + o + 1$ is redundant. Once (N2) is recovered, (N1) implies

$$V < B(F + 1), \qquad V \geq n^2 > F(F + 1), \qquad B > F \geq 2,$$

and consequently

$$V < B(F + 1) \leq B^2 \leq N \leq q. \tag{N5}$$

Thus $o = q - B - 1 \geq 0$ is restored *before* the factorial digit is interpreted. Since every $\binom{n}{j}$, $0 \leq j \leq t$, is at most $n^t = V < q$, that digit gives $B = \binom{n}{t} > 0$. The descending floor theorem, valid for $t \geq 3$, then gives $F = t!$. For completeness, the canonical ordinary low coefficients are at most S^x, those for J are also at most S^x once $\ell(t+1) < x$, and factorial low coefficients are at most $V < N$. Every group coefficient sum is therefore at most

$$5S^x + N < N^x S^x = q$$

for $N \geq 4, x \geq 2, S \geq 2$. Its geometric low-digit expansion is at most $D - 1$, supplying the natural pooled slack in (N4).

3.2 A new even-horizon cofinal beta specialization

The new structural step sets

$$t = 2t_0 + 4, \qquad y + 1 = V, \tag{N6}$$

in place of $t = t_0 + 12$ and $y = F + r_0 + 1$, deleting r_0. The finite coefficient change substitutes $t_{0,\text{old}} = 2t_{0,\text{new}} - 8$ in every raw residual. **This is not a natural-coordinate bijection.** Its domain theorem is the soundness and constructive completeness argument that follows.

Acyclic soundness. The power-modulus row still gives $F \geq 2, n \geq 3$, and the first endpoint $x = F + 1 + (y+1)u_0$ gives $x \geq F + 1 \geq 3$. The exponents $t, x, y+1$ are positive from the natural domains alone. Thus the frame, restored full cap and unchanged numerator incidences recover (N2) without first needing $y \geq F + 1$. Afterwards

$$y = (F^2 - 1)^t - 1 \geq F + 1.$$

The last endpoint $x = 1 + (1 + (t+1)y)u_t$ has $u_t \geq 1$, so

$$x \geq 2 + (t+1)y > \max(t+1, y+1). \tag{N7}$$

All required denominator divisibilities for the common-root modulus Nq^x follow before the first binomial decoding. The bootstrap (N5) then identifies $F = t!$.

Now $t \geq 4, F \geq 24, n \geq 575$, and hence

$$y \geq 575^4 - 1 > 2^{36}, \qquad x > 2^{38}.$$

With $R = 2 + t + x + y + F$, one has $R < 2x$. The unchanged source ledger has $E = 2^{23}$; for every integer $x \geq 2^{38}$,

$$(2x)^E < 2^x.$$

The base case is $39E < 2^{38}$; induction uses $(1 + 1/x)^E < 2$ for $x \geq 2E$, obtained by majorizing the binomial expansion by the geometric series in E/x. The label ledger has $\ell = 2^{22}$, and $y > \ell$ gives $\ell(t+1) < (t+1)y < x$. These replace exactly the old horizon-12 majorant and label estimates. They justify all range digits and the strict local bound below 2^x.

Constructive cofinal completeness. Expand a finite refutation to a finite proof tree. Choose an *even* $t \geq 4$ at least its rule occurrences, parameters, full relevant label sums, and truth-padding label sum, before defining $F = t!$. At every backward expansion stage, each pending nontruth vertex is the root of a distinct unexpanded subtree. There are therefore at most t nontruth factors. Its label $j \leq t$ has weight $1 + jF \leq 1 + tF < F^2 - 1$; truth weights are one. Every pending product obeys

$$S_i \leq (1 + tF)^t < (F^2 - 1)^t = V,$$

so $S_i \leq y = V - 1$. Also $y \geq F + 1$. Since $F^2 - 1 \equiv -1 \pmod{F}$ and t is even, $F \mid y$. The beta moduli $1 + (i+1)y$, $0 \leq i \leq t$, are pairwise coprime: a common prime would divide an index difference at most t while remaining coprime to y, contradicting $t! \mid y$. CRT supplies x with exactly the prescribed pending-state residues, including $F+1$ initially and one finally. Its Euclidean quotients and all endpoint witnesses are natural.

Continue in this order: descending reciprocal moduli $d_i = F^2/i - 1$, their product B, $N = B^x$, precision-N^2 CRT representatives with the original one-representative cofinal lift, then S, q, denominators, frame and lower Pell witness, a sufficiently large common-root-compatible A, principal and seed pairs, and fresh positive-center auxiliary/comparison witnesses. The original semantic anchor dominance follows from that unchanged representative lift. None of these later choices changes the earlier beta parameters. The pooled slack construction above completes the new graph. This proves cofinal completeness rather than merely a one-way restriction of witnesses.

3.3 Exact compilation, verification, and selected tradeoffs

The first composed source has 325 variables and 276 rows; (N6) gives 324 variables and 276 rows. Certified natural contractions yield the controls

	variables	rows	terms	height
cofinal dimension	219	171	9,253	560,710,452
cofinal sparse	221	173	6,269	83,847,181
noncofinal balanced	221	172	6,561	83,878,986

Their final residual bases use respectively 32, 44 and 41 elementary integer shears $R_i \leftarrow R_i + kR_j$, $i \neq j$. Applying the opposite shears in reverse order recovers every original residual exactly. No coordinate changes domain or sign. The final metrics are the table at the beginning of this section.

The independent source checker reconstructs every raw mutation and the cofinal coefficient translation. The natural contraction checker expands each rational sum-of-affine-squares certificate plus nonnegative-monomial remainder, replays every substitution, and checks the full collected SOS. The residual-basis checker verifies forward and inverse maps, refreshed embedded metrics, every final coefficient, and deliberate corruptions. Every rendered TeX residual is parsed back. Normal and optimized Python rebuild the selected files byte-identically, including the complete retained offline Round18 source/compiler closure.

The literal equations are in `natural219/`, `sparse221/`, and `balanced221/`; the corresponding pre-basis controls are `dimension219/`, `cofinal221/`, and `projected221/`. The ordinary proofs are `PROOF.md` and `COFINAL_PROOF.md`. The retained C222/ZF/ZFC interfaces are inputs, not fresh end-to-end proof-assistant certificates.

Finally, full common-denominator linear normalization was tested but not selected: its raw count falls from 325 to 323, yet its contracted count rises from 221 to 231. This six-power natural architecture uses direct Pell numerators $Y(q+1) - Z$, and multiplying their wires by denominator ratios blocks earlier quadratic monic eliminations. The result is a measured limitation of this implementation, not a general obstruction.

4 Ten-positive equations: projection before full denominator recovery

The two useful new local operation/degree points are **79/382** and **80/338**, with ten independently strictly positive integer unknowns. The whole indexed BFS943 radix67 coefficient tuple is retained, including all four complete payloads and their finite reconstruction. Its largest port has 592,884 bits. Expanded polynomial coefficients can be larger.

Chart	Transport weight	Binary operations	Exact degree
$X = qw,\ Y = q^2 s,\ D = q^5$	V^2	78	516
$X = qw,\ Y = q^2 s,\ D = q^5$	bV	79	382
$X = q^2 w,\ Y = qs,\ D = q^6$	V^2	79	456
$X = q^2 w,\ Y = qs,\ D = q^6$	bV	80	338

The square-weight rows are dominated controls, not frontier claims. The incoming *Centered Residues* package already supplies 75/626, 76/464 and 77/460; its XY, kY arithmetic sharing is inherited here. The incoming *Coprime Products* package already proves the q^2 chart at 80/382; composing it with that sharing saves one operation. The substantive new chart is $Y = qs, X = q^2 w, D = q^6$, with projection before full-denominator recovery. The older rejected chart $Y = qs, X = qw$ had

only q^3 visibly in its first recurrence modulus. It is not reasserted. Sign-free unshifting, the two signed third-index exclusions, Pell doubling, and the native calibration are credited in the copied proof sources. No historical priority or global optimum is claimed.

Theorem and boundary. At the explicitly retained ordinary Pell/native/refutation calibration, each polynomial below has a positive zero if and only if the inherited syntactic calculus has a finite ZFC refutation. Equivalently, ZFC proves that absence of a positive zero is equivalent to Con(ZFC). This neither proves consistency nor asserts that a zero exists. It is ordinary mathematics formalizable in ZFC, not a new end-to-end Lean/Rocq derivation or emitted object-ZFC proof. The finite verifier checks algebra and artifacts, not the existence or absence of refutations.

4.1 Complete polynomial definitions

The ten inputs are

$$J, F, \alpha, i, T, w, s, n, \eta, \zeta \in \mathbb{Z}_{>0}.$$

The fixed ports are the entire calibrated tuple β, K_0, M_C, M_F, with $\beta \geq 31$ odd, $K_0 > 0$, and $0 < M_C < \beta < M_F < 2\beta$. Choose one complete chart:

	W_0	X	Y	D	σ (proof only)
q^2	w	qw	$q^2 s$	q^5	s/q
q^1	qw	$q^2 w$	qs	q^6	s/q^2

In both cases $X = qW_0$ and $\sigma = Y/q^3$. Define

$$\begin{aligned}
h &= \beta J, & q &= h + 1, & C &= q - F - \alpha,\\
k &= \eta + \zeta, & c &= kY + \eta, & A &= Y(X+1) + 1,\\
\Delta &= A^2 - 1, & H &= 4A - 5, & v &= XY^2,\\
\ell &= (XY)(kY), & E &= 1 + \ell(\ell + k), & & \\
e &= k - R - 1, & B &= (K_0 + W_0)C - F, & & \\
U &= ic^2, & a &= \Delta U^2 E, & F_0 &= 1 + a,\\
G &= EF_0, & \mathcal{F} &= 1 + 4aF_0, & Q &= \Delta(\mathcal{F} - 1),\\
b &= cT - R, & V &= \mathcal{F}b - c, & K &= X(3c - 2X) + 2.
\end{aligned}$$

Also set

$$R = [h(q - F) + \alpha](q^2 - 1) + (M_C + qM_F)J.$$

Choose $L = V^2$ or $L = bV$. The final polynomial is

$$\begin{aligned}
M &= hDH,\\
Z &= hD[H(n - LB) + K] - H[DB + h(e - sq^4)], &(1)\\
P &= Q(Z^2 - M^2 G V^2) + (M^2 G - Z^2). &(2)
\end{aligned}$$

All symbols are polynomial abbreviations, not additional unknowns. Every square and scalar multiplication is charged as a binary gate; there is no root, division, or exponentiation instruction in the four emitted DAGs. For proof only set $z = Z/M$, $\kappa = K/H$, $\rho = e/D$, $t = B/h$, and $\mathcal{W} = 1 + hL$. Since $D\sigma = sq^4$ in both charts,

$$z = n + \kappa + \sigma - \mathcal{W}t - \rho. \tag{3}$$

None of these quotients is initially presumed integral.

4.2 Roots, folding, and unconditional bounds

Positive inputs imply $q \geq 32$, $X, Y \geq q$, $q \mid X, Y$, $k \geq 2$, $c > k$, $A > 4$, $\Delta \geq 3$, $U \geq c^2$, and $E > 1$. Thus $a > c$, $\mathcal{F} > 2c$, and $Q > c^2$. For any integral b, $V \neq 0$: if $b = 0$ then $V = -c$; otherwise $|V| \geq \mathcal{F} - c > 1$. Identically,

$$\mathcal{F} = (1 + 2a)^2, \qquad Q = 4\Delta^2 U^2 G, \qquad \gcd(E, F_0) = 1.$$

At $P = 0$,

$$(Q - 1)z^2 = G(QV^2 - 1).$$

For every prime dividing G, the two other integer factors are -1 modulo that prime. Rational-square valuations force $G = g^2$, $g > 0$ integral. Coprimality gives $E = \tau^2$, $F_0 = f_0^2$, $g = \tau f_0$. The positive rational $y = |z|/g$ satisfies

$$QV^2 - (Q - 1)y^2 = 1. \tag{4}$$

It is integral by adjacent-Pell descent: for reduced $y = r/m$, one has $m^2 \mid Q - 1$, and the norm-preserving map

$$v' = (2Q - 1)v - 2(Q - 1)y, \qquad y' = (2Q - 1)y - 2Qv$$

from $v = |V| > 1$ has nonzero integral v', $|v'| < v$, and unchanged reduced denominator of y. Sign reflection and descent reach $v = y = 1$. Hence $z = \varepsilon g y$ is integral, $\varepsilon \in \{1, -1\}$.

The proof integers $f = 1 + 2a$ and $I = 2Ug$ obey

$$f^2 = \mathcal{F}, \quad f^2 - \Delta I^2 = 1, \quad c^2 \mid I, \quad Q = (\Delta I)^2.$$

The retained signed all-index folding theorem applies to $c \leq I$ and $V \equiv -c \pmod{f}$, without a sign assumption on V. It gives $c = \psi_A(p)$ for odd positive p. The first norm gives $k = 2\psi_{2XY^2+1}(n_0)$, $n_0 \geq 1$. At $n_0 = 1$, $k = 2$ and $c = 2Y + 1$ lies strictly between $\psi_A(1) = 1$ and $\psi_A(2) = 2A$. Consequently

$$n_0 \geq 2, \quad k \geq 8XY^2 + 4, \quad c > 8XY^3 \geq 8q^5, \quad c > X, \quad K > 0. \tag{5}$$

No positivity of B, C, R, b has been assumed.

4.3 Unshifted transport, including both signed exclusions

If $b = 0$, equation (4) gives $y^2 = c^2 + (c^2 - 1)/(Q - 1)$ strictly between consecutive integers. Thus $b \neq 0$, $\operatorname{sign} V = \operatorname{sign} b$, and $|V| > |b|$. Directly,

$$a^2 - g^2 = E[\Delta U^2 E(\Delta U^2 - 1) - 1] > 0, \quad \mathcal{F} - c > 3a > 3g, \quad y < \tfrac{3}{2}|V|.$$

Hence $gy < V^2/2$. For $L = bV$, suppose first that the main index is $p = 3$. Then $c = 4A^2 - 1$. If $n_0 \geq 3$, then $kY > 32X^2Y^5$, whereas $A < 2XY$ gives $c < 16X^2Y^2$, impossible. Therefore $n_0 = 2$, $k = 8XY^2 + 4$. With $u = X + 2 - 2Y$,

$$\eta = 4Y^2(Xu + 1) + 8Y(X + 1) + 3 - 4Y.$$

For $u \leq -1$ this is negative, since $Y(X - 1) > 2(X + 1)$; for $u \geq 2$ it exceeds k. Thus $u = 0$ or 1, giving $2Y - X \in \{1, 2\}$, contrary to $q \mid X, Y$, $q \geq 32$. Hence $p \geq 5$ and $c > c_3 := 4\Delta + 3$.

Put $S = \Delta I$, so $Q = S^2$. Pell classification of (4) gives $|V| = \chi_S(j)/S$ for odd j; even traces are ± 1 modulo S. The first index gives $|V| = 1$. At the third index $|V| = 4Q - 3$, and the two signs would imply

$$\mathcal{F}(b - 4\Delta) = c - c_3 \quad \text{or} \quad \mathcal{F}(b + 4\Delta) = c + c_3.$$

Both are excluded by $0 < c - c_3 < c + c_3 < 2c < \mathcal{F}$. Therefore $j \geq 5$, $|V| \geq 16Q^2 - 20Q + 5 > 8Q^2$, and

$$|b| > \frac{8Q^2 - c}{\mathcal{F}} > 8\mathcal{F} - \tfrac{1}{2} > 4\mathcal{F} > 4g.$$

It follows that $L = bV > 0$ and $gy < 3L/8 < L/2$.

For either weight $e < b$, since $b - e = cT - k + 1 > 0$, so $\rho < |b|/D$. If $B \leq -1$, then $\mathcal{W}t < -L$; using $\kappa, \sigma > 0$ in (3) yields

$$n = \varepsilon gy - \kappa - \sigma + \mathcal{W}t + \rho < L/2 - L + |b|/D < 0,$$

since $D > 2$. This contradicts the actual positive input n. Thus $B \geq 0$, hence $C > 0$. Only now use the mask bound $3q < R < q^4 - 2$. Equation (5) gives $c > 2R$, so $b, V > 0$. The retained normalized-rank and signed auxiliary-index lemmas, with $c^2 \mid I$, $c > 2R, c > 2p$, and the fixed-minus residues, give $p = R$. Mask parity yields even q and odd J.

4.4 Authenticate the first index before the full denominator

Integral z in (3) implies

$$D \mid hH(e - sq^4).$$

Since $H = 4Y(X + 1) - 1 \equiv -1 \pmod{q}$ and $h = q - 1$, $\gcd(q, hH) = 1$. As $D = q^5$ or q^6, this implies $q^4 \mid e$. The universal first recurrence is

$$k \equiv 2n_0 \pmod{4XY^2}.$$

Its modulus is visibly divisible by q^4: $XY^2 = q^5ws^2$ in the q^2 chart, and $XY^2 = q^4ws^2$ in the q^1 chart. Therefore $q^4 \mid 2n_0 - R - 1$. Moreover $P_0 = 2XY^2 + 1 > A$, because $P_0 - A = Y(2XY - X - 1) > 0$. If $n_0 \geq R$, monotonicity in base and index gives $k/2 \geq \psi_A(R) = c$, contrary to $c > k$. Hence

$$1 \leq n_0 < R, \qquad |2n_0 - R - 1| < R < q^4,$$

and necessarily $2n_0 = R + 1$. Returning to the full recurrence gives

$$4XY^2 \mid e. \tag{6}$$

In the q^2 chart this already gives $q^5 \mid e$. In the q^1 chart it does *not* yet give $q^6 \mid e$. That conclusion is deliberately deferred.

4.5 Projection first, full denominator second

Modulo H, the literal numerator gives $Z \equiv hDK$. Since $H \mid M$ and z is integral, $H \mid hDK$. As D is a power of q and $H \equiv -1 \pmod{q}$, cancellation is legitimate here and gives

$$H \mid hK, \tag{7}$$

without assuming ρ integral. This is the new ordering.

Write $R = 2r + 1$; $r \geq 48$. The retained scale-one ratio lemma gives

$$\xi < \frac{\psi_A(2r+1)}{\psi_{2XY^2+1}(r+1)} < \xi\left(1 + \frac{1}{Y(X+1)}\right)^{2r}, \qquad \xi = \frac{(X+1)^{2r}}{X^r}.$$

Together with $Y < c/k < Y + 1$, this implies $Y > X^r/3$ and $X^{r+1} < 3Y(X+1) < H$. The universal projection identity, odd H, and (7) give $H \mid h(X - 2^R)(2^R X + 1)$, hence also divisibility after multiplication by q^3. Here is an explicit whole-product bound. Put $t_0 = 2^R = 2 \cdot 4^r$. Since $q^3 h < X^4$,

$$\begin{aligned} q^3 h\,|(X - t_0)(t_0 X + 1)| &< X^4(X + t_0)(t_0 X + 1) \\ &\leq 4 \cdot 4^r X^6 + 8 \cdot 16^r X^5 < X^{r+1}. \end{aligned}$$

After division by X^{r+1} the two terms decrease on $X \geq 32, r \geq 48$; at $(32, 48)$ their sum is $2^{-117} + 2^{-25} < 1$. Thus the entire product is zero, and its positive factors give $X = 2^R$. The universal identity then makes $\kappa = K/H$ a positive integer. No arbitrary factor of a composite modulus was canceled.

Since $q \mid X = 2^R$ and $q \geq 32$, $q = 2^m$ for $m \geq 5$. The induction $2^m \geq 2m$ for $m \geq 2$ yields

$$R > 3q = 3 \cdot 2^m \geq 6m, \qquad q^6 \mid X.$$

Equation (6) now gives $q^6 \mid e$ in the new chart. Thus $\rho = e/D$ is integral in both charts. This is a property of proved zeros, not a claim that $q^6 \mid 4XY^2$ for all positive inputs.

Equation (3) now makes $s/q^j - \mathcal{W}B/h$ integral, where $j = 1$ or 2. Clear $q^j h$ and reduce first modulo q^j, then modulo h. Since $\gcd(q, h) = 1$ and $\mathcal{W} \equiv 1 \pmod{h}$,

$$q^j \mid s, \qquad h \mid B, \qquad q^3 \mid Y, \qquad t = B/h \in \mathbb{Z}_{\geq 0}.$$

The missing native divisibility has been recovered, not assumed.

4.6 Native soundness and constructive positive completeness

Soundness now supplies positive outer data J, F, α, W_0, Y with $C > 0$, $X = qW_0 = 2^R$, $q^3 \mid Y$, and

$$(K_0 + W_0)C - F = t(q - 1), \qquad t \geq 0.$$

The retained fixed-minus parity gives $R \equiv 3 \pmod 4$; the ratio and exact half-binomial rounding theorem identifies the native Y. Reconstruct fresh source witnesses at $A_{\text{source}} = Y(X+1) + 2$, *not* our A. With $z_{\text{source}} = t + 1 > 0$, the native transport identity is exactly

$$(K_0 + W_0)C + (q - F) - z_{\text{source}}(q - 1) = 1.$$

The whole indexed BFS943 calibration yields a finite syntactic ZFC refutation. The scalar port inequalities used early are not a substitute for that decoder.

Conversely a finite refutation gives canonical data $q = 2^m$, $X = 2^R = qW_{\text{native}}$, $R > 3q$, $R \equiv 3 \pmod 4$, $q^3 \mid Y$, and the same transport $t \geq 0$. The preceding elementary bound gives $q^6 \mid X$. Choose $w = X/q$, $s = Y/q^2$ for the q^2 chart; choose $w = X/q^2$, $s = Y/q$ for the q^1 chart. All are positive integers, and in either case $W_0 = W_{\text{native}}$: the coefficient in the native transport is preserved. Set

$$c = \psi_A(R), \quad k = 2\psi_{P_0}((R+1)/2), \quad \tau = \chi_{P_0}((R+1)/2), \quad \eta = c - kY, \quad \zeta = k - \eta.$$

The retained exact ratio/rounding theorem gives $0 < \eta < k$. The full recurrence and $q^6 \mid X$ give integral $\rho \geq 0$; projection gives integral $0 < \kappa < Xc$.

A positive multiple of the Pell matrix period modulo $c^2\tau$ supplies a seed pair (f_0, I_0) with $c^2\tau \mid I_0$. Take $i = I_0/(c^2\tau) > 0$. Then $a = \Delta I_0^2 = f_0^2 - 1$, $F_0 = f_0^2$, $G = (\tau f_0)^2$, and the doubled pair is exactly $f = 2f_0^2 - 1$, $I = 2f_0 I_0$. The retained auxiliary-completeness lemma supplies positive V, y and the positive integer

$$T = \frac{V + c + Rf^2}{cf^2}.$$

This is the actual unshifted T, not a possibly negative inverse shift. For either positive weight, choose

$$n = gy + (1 + hL)t + \rho - \kappa - \sigma.$$

All terms are integral since canonical $\sigma = Y/q^3$ is integral. Also $\sigma < Y < Xc$, $\kappa < Xc$, $I_0 \geq c^2$, and $f_0 > (A-1)I_0 > 2Xc$, so $g = \tau f_0 > \kappa + \sigma$. Thus $n > 0$ and its carrier is $z = gy$, proving $P = 0$. Every independent coordinate is positive. This is a fresh canonical witness construction, not an unproved positive inverse shear.

4.7 Exact global degrees and reproducibility

The literal sharing is $A - 1 = XY + Y$ and $\ell = (XY)(kY)$. From the incoming q^4 charts, remove the private q^3 multiplication, change Y, add $D = q^4 q$ or $q^4 q^2$, and add sq^4 and $e - sq^4$. This is net two operations in the q^2 chart. The q^1 chart additionally splits X into $W_0 = qw$, $X = qW_0$, using W_0 in B. All gates, inputs, and four fixed ports are live.

Put $q_0 = \beta J$, $k_0 = \eta + \zeta$, and $C_0 = q_0 - F - \alpha$. The complete leading forms for the q^2 chart are

$$[P_{V^2}]_{516} = 2^{14} q_0^{272} w^{46} s^{108} i^{20} k_0^{64} T^4 C_0^2,$$
$$[P_{bV}]_{382} = -2^{10} q_0^{206} w^{34} s^{80} i^{14} k_0^{46} T^2.$$

In the q^1 chart, $\deg(cT) = \deg R = 4$; dropping R would give an incorrect leading form. Define

$$B_0 = k_0 sT - q_0^2(q_0 - F).$$

Then the complete forms are

$$[P_{V^2}]_{456} = 2^{14} q_0^{212} w^{46} s^{104} i^{20} k_0^{60} B_0^4 C_0^2,$$
$$[P_{bV}]_{338} = -2^{10} q_0^{162} w^{34} s^{78} i^{14} k_0^{44} B_0^2.$$

Their sparse expansions have respectively $390, 47, 3750, 274$ monomials. The selected maximal-T coefficients are $2^{14}\beta^{272}$, $-2^{10}\beta^{206}$, $2^{14}\beta^{212}$, $-2^{10}\beta^{162}$, nonzero at the actual $\beta > 0$. These prove exact multivariate degrees, not numerical-ray estimates. The unique top summand is QZ^2 for V^2 and $-QM^2GV^2$ for bV.

The independent standard-library checker does not import the producer. It covers every actual row with 28 free-polynomial cuts per graph, proves the entire leading forms, checks liveness and the whole tuple, reconstructs every coefficient byte, and rejects ten genuine algebraic corruptions per graph. Normal and optimized Python receipts are identical. The complete unbounded proof is also supplied in `PROOF.md`; the active local `SOURCE_PINS.json` authenticates all copied ordinary dependencies.

```
python3 -B build.py --check
python3 -B verify.py --output /tmp/positive-check.json
python3 -B -O verify.py --output /tmp/positive-check-O.json
python3 -B rebuild_coefficients.py
```

All source intake was read-only. No remote repository was modified.

5 Complete fixed-coefficient recipe

The following finite recipe is retained from the inspected ProveIt indexed constant specification, whose exact source and SHA-256 are recorded in `provenance/indexed_constants.tex` and `SOURCE_PINS.json`. It defines the entire unchanged BFS943 tuple, not a new specialization. The symbol K in the inherited recipe is renamed K_0 to agree with Section 4; symbols in this section are local fixed data, not additional unknowns. The supplied standard-library reconstruction checks every resulting byte against all four literal circuits. The source shift is already included in M_F.

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{67}, \quad B = \rho^{8807}, \quad \beta = B - 1, \quad G_m(x) = \sum_{j=0}^{m-1} x^j.$$

Define

$$\begin{aligned}\mathcal{E} = {} & \{0, \dots, 942\} \cup \{983, \dots, 992\} \cup \{1003, \dots, 1011\} \\ & \cup \{1013, \dots, 1031\} \cup \{1063, 1064, 1095, 2195, 3296\}.\end{aligned}$$

Then

$$\begin{aligned}M_C &= B - 1 - \sum_{e \in \mathcal{E} \setminus \{1\}} \rho^e - 2\rho^{1063}, \\ M_F &= B - 1 + 4 + (2^{51} - 2)(\rho^{4393} + \rho^{5495}) \\ &\quad + \rho^{1013} G_{10}(\rho) + \rho^{1065} G_9(\rho) + \rho^{2195} + \rho^{3296} \\ &\quad + (2^{65} - 1)\rho^{8792} G_{13}(\rho) + (2^{19} - 1)\rho^{8805}, \\ K_0 &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\ &\quad + (\rho^{4393} + \rho^{5495})\Phi(1/\rho),\end{aligned}$$

where

$$\begin{aligned}\Phi(x) &= (1 + 2^{51}) G_{943}(x) - 2^{10} \sum_{i=0}^{942} v(t_i) x^i \\ &\quad + \sum_{(b,e,m) \in \mathcal{C}} 2^b x^e G_m(2x), \\ \mathcal{C} = {} & \{(10, 1023, 8), (18, 1013, 9), (27, 1003, 8), \\ & (35, 983, 9), (44, 1031, 1), (45, 1022, 1), (46, 1011, 1), \\ & (47, 992, 1), (48, 1095, 1), (49, 2195, 1), (50, 3296, 1)\}.\end{aligned}$$

The rational intermediate in K_0 cancels exactly; all four results are positive integers.

The 943 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. Let

$$h(q,b) = 2(q+1)+b, \quad \mathcal{R} = \{u_j : d_j = 1,\ 0 \le j < 430\},$$
$$\mathcal{L} = \{u_j : d_j = 0,\ 0 \le j < 430\},$$
$$r_q = 6 + |\mathcal{R} \cap \{0,\dots,q-1\}|, \quad l_q = 205 + |\mathcal{L} \cap \{0,\dots,q-1\}|.$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 942$; all index ranges are fixed:

index i	tile $t_i = (w,n,e,s)$
$0, 1$	$(2,435,3,1),\ (5,0,0,0)$
$j+2\ (j = 2q+b < 430)$	$((1-d_j)l_{u_j}, h(q,b), d_j r_{u_j}, v_j)$
$432+b\ (b=0,1)$	$(0, h(215,b), 0, h(215,b))$
$434+b\ (b=0,1)$	$(0,b,0,b)$
$424+2r_q+b\ (q\in\mathcal{R}, b=0,1)$	$(r_q, b, 0, h(q,b))$
$424+2l_q+b\ (q\in\mathcal{L}, b=0,1)$	$(0, b, l_q, h(q,b))$
$924+j\ (0 \le j < 4)$	$(0, j+430\lfloor j/2\rfloor, 0, 438)$
928	$(2,434,2,0)$
929	$(3,436,4,4)$
930	$(4,436,4,0)$
$497+t\ (t=434,435,436)$	$(1,438,1,t)$
$934+3a+b-6\lfloor a/4\rfloor$ $a\in\{0,1,4\},\ b=0,1,2$	$(a,437,b,437)$

For $t = (w,n,e,s)$ put

$$v(t) = w + 2^8 n + 2^{17} e + 2^{25} s + 127\,2^{34}.$$

A closed decoder for the 430 actions. Put $D_b(A,k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b\rfloor$. For $0 \le q < 215$, $b = 0,1$, $j = 2q+b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \left\lfloor \mathcal{A}/2^{636} \right\rfloor \bmod 2^{1675},$$
$$E_0 = \left\lfloor \mathcal{A}/2^{2311} \right\rfloor,$$
$$\nu_j = D_2(N,j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430+q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14(1-m_q) + m_q D_{16}(E_0, \tau_q),$$
$$u_j = \nu_j(1+\sigma_j) + (1-\nu_j) D_{216}(T_0, j-\sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b+1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1,1,1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0,\dots,61$, is

```
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz.
```

Concatenate the following 436 characters as digits $e_0,\dots,e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
```

```
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSD0Tgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```


6 Centered integer-root charts: fewer operations at fixed degree

A polynomial change of the carrier coordinate lowers the literal arithmetic length of two retained fixed equations. The new points are

14 integers, deg 88, 110 operations		13 integers, deg 156, 117 operations

The first saves five operations over 14/88/115 and strictly dominates the previous 14/94/110 alternative. The 14/86/112 point remains a genuine lower-degree tradeoff. The second saves eight operations over 13/156/125. The six complete fixed numerals and the ordinary native/refutation calibration are unchanged. These are not global optimality or new kernel-certificate claims.

6.1 The complete arithmetic interface

The fixed ports are $(\beta, K_0, M_C, M_F, A_c, B_c)$, with M_F already shifted, $\beta = 2^{1064921} - 1$, $A_c = 2^{1113}$ and $B_c = 2^{666104}$. All following letters are polynomial abbreviations, not additional witnesses:

$$
\begin{gathered}
J = 6j + 1, \quad q_m = \beta J, \quad q = q_m + 1, \quad W = 12w + 4, \\
X = qW, \quad Y = v, \quad F = (K_0 + W)C - zq_m, \\
R = (q(q - F) - C)(q^2 - 1) + (M_C + qM_F)J, \\
E = XY, \quad a = E + Y, \quad H = 4a + 3, \quad \Delta = a^2 + H, \\
k = R + 1 + hE, \quad \eta = c - Yk, \quad L = E(Yk), \quad I = (2r + 1)c^2, \\
M = 2q^3, \quad U = MH, \quad K_\ell = X(2X - 3c) - 2, \\
Z = Un + MK_\ell + H(Y - q^3), \quad D_t = 1 + L(L + k), \quad b = 2L + k, \\
D_e = (A_cF - q)^2 + (B_cC - q)^2, \\
\mathcal{B} = (Yk)^2(q^2 - D_e) - q^2(4(Y\eta)^2 + (kq^3)^2).
\end{gathered}
$$

The retained fifteen-integer parent additionally has f, y, p_1, p_2, p_3 and

$$
\begin{aligned}
Q &= \Delta(f^2 - 1), & V &= f^2(cT - R) - c, \\
A &= (Z + Uk)^2 + 4U(LZ - U), & B_s &= f^2 - \Delta I^2 - 1, \\
C_a &= V^2 - (Q - 1)y(2V + y) - 1, \\
B_g &= \mathcal{B} - 1 - p_1^2 - p_2^2 - p_3(p_3 + 1).
\end{aligned}
$$

Its single polynomial is $A^2 + B_s^2 + C_a^2 + B_g^2$. The complete ordinary proof and source equation are retained under `roots/inherited/baseline/source/`. The exact identity central to the new chart is

$$A = (Z + Ub)^2 - 4U^2D_t. \tag{8}$$

Before imposing any equation, q is even and nonzero, H is odd and nonzero, hence $U \neq 0$. Also $|X| \geq 8$, and $a = (X + 1)Y$ is either zero or has $|a| \geq 7$, so $\Delta = (a + 2)^2 - 1 \geq 3$. With $D = XY^2$, we have $D_t = 1 + D(D + 1)k^2 \geq 1$. Since D is even, D_t is odd. Finally $\Delta \equiv 0$ or $3 \pmod 4$, so $B_s \not\equiv 2 \pmod 4$ and $B_s \neq -2$.

6.2 The fourteen-variable equation and all-zero maps

Use the unrestricted inputs

$$(j, C, z, h, r, v, w, T, c, n, f, y, p_2, p_3).$$

Keep Q, V above and define

$$\begin{aligned} S &= f^2 - \Delta I^2, & E_a &= V^2 - (Q-1)y(2V+y), \\ G &= \mathcal{B} - 1 - p_2^2 - p_3(p_3+1), & & \\ g_a &= U^2 G, & d_b &= 4U^2 D_t, \\ N_2 &= (Z^2 - g_a - d_b)^2 - 4g_a d_b, & & \\ P_{14} &= (N_2+1)SE_a - 1. & & \end{aligned}$$

There are no division or radical gates. Since N_2 is a square modulo 4 and $S = B_s + 1 \neq -1$, the three integral factors in $P_{14}+1$ can multiply to 1 only when each is 1. Thus $P_{14} = 0$ is equivalent to $N_2 = B_s = C_a = 0$.

For soundness write $\rho = Z/U \in \mathbb{Q}$ and $A_r = 4D_t \geq 4$. Then

$$(\rho^2 - A_r - G)^2 - 4A_r G = 0. \tag{9}$$

Negative G is impossible because the left side would be a square plus a strictly positive number. Hence $G \geq 0$ and $\mathcal{B} > 0$, forcing $Yk \neq 0$. As $\mathcal{B} \leq Y^2k^2q^2$ and $|D| \geq 8$,

$$A_r \geq 4 + 2X^2Y^4k^2 \geq 4 + 32q^2Y^4k^2 > Y^2k^2q^2 \geq \mathcal{B} > G.$$

Consequently $\rho \neq 0$: otherwise (9) says $A_r = G$. Only now form

$$p = \frac{\rho^2 + G - A_r}{2\rho}, \qquad u = \rho - p.$$

Their squares are G and A_r. A rational number with integral square is integral, by reduction to lowest terms. Therefore $p, u \in \mathbb{Z}$, and u is even. Write $u = 2\tau$. The branch $G = 0$ is included: it gives $p = 0$, not a discarded solution. Restore

$$p_{1,\mathrm{old}} = p, \qquad n_{\mathrm{old}} = n - b - p.$$

Then $Z_{\mathrm{old}}/U + b = 2\tau$, so (8) gives $A_{\mathrm{old}} = 0$. The strong and auxiliary rows are unchanged, and $p^2 = G$ restores the guard.

Conversely, at any parent zero, $(Z_{\mathrm{old}}/U + b)^2 = 4D_t$ makes that rational number an even integer 2τ. Delete p_1 and set

$$n_{\mathrm{new}} = n_{\mathrm{old}} + b + p_{1,\mathrm{old}}.$$

Now $\rho = 2\tau + p_{1,\mathrm{old}}$ and $G = p_{1,\mathrm{old}}^2$; substitution proves $N_2 = 0$, while the other rows are unchanged. These maps apply to every parent zero, not only a canonical family.

6.3 The thirteen-variable centered norm

Use

$$(j, C, z, h, r, v, w, T, c, n, y, p_1, p_2),$$

and set

$$\begin{aligned}
D_s &= \Delta I^2, & F_s &= 1 + D_s, & Q &= \Delta D_s,\\
V &= F_s(cT - R) - c, & E_a &= V^2 - (Q-1)y(2V+y), & &\\
G &= 4(\mathcal{B} - p_1^2 - p_2^2) - 3, & & & &\\
g_a &= U^2 G, & d_b &= 4U^2 D_t, & f_c &= 16U^2 F_s,\\
g_{ad} &= g_a d_b, & g_{ab} &= g_a + d_b, & Z_2 &= Z^2,\\
K &= (Z_2 - g_{ab} - f_c)^2 - 4(g_{ad} + g_{ab} f_c), & & & &\\
N_3 &= K^2 - 64 g_{ad} f_c Z_2, & P_{13} &= (N_3+1)E_a - 1. & &
\end{aligned}$$

Because N_3 is a square modulo 4, $P_{13} = 0$ is exactly $N_3 = 0$ and $E_a = 1$. The complete signed factorization is

$$\frac{N_3}{U^8} = \prod_{\varepsilon \in \{\pm 1\}^3} \left(\rho - \varepsilon_1\sqrt{G} - 2\varepsilon_2\sqrt{D_t} - 4\varepsilon_3\sqrt{F_s}\right), \qquad \rho = \frac{Z}{U}. \tag{10}$$

This is a polynomial identity valid over $\mathbb{C}$. Here $D_t, F_s \geq 1$ and $G \equiv 1 \pmod 4$. If $G < 0$, each factor has a nonzero imaginary part from its sole imaginary radical; thus a zero forces $G > 0$. The case $G = 0$ is impossible from the outset, and neither other radicand can vanish.

To handle shared square classes, write each positive radicand as $u_i^2 d_i$ with $u_i > 0$ integral and d_i squarefree. A vanishing factor gives

$$\rho = \varepsilon_1 u_1\sqrt{d_1} + 2\varepsilon_2 u_2\sqrt{d_2} + 4\varepsilon_3 u_3\sqrt{d_3}.$$

Distinct squarefree radicals are rationally independent, as follows by character averaging over the prime-root sign-change automorphisms of a multiquadratic field. Grouping a common irrational class cannot give zero: a group containing the G term has an odd coefficient and all other coefficients even; a group without G but containing D_t has one coefficient of exact 2-adic valuation 1 and at most a multiple of 4; a group containing only F_s has one nonzero coefficient. The oddness of G, D_t supplies the stated valuations; F_s need not be odd. An even squarefree class can only contain F_s and is likewise impossible.

Thus all roots are integers. Select their signs from the zero factor and write

$$g^2 = G, \qquad \tau^2 = D_t, \qquad f^2 = F_s, \qquad \rho = g + 2\tau + 4f.$$

The integer g is odd. In particular ρ is odd and nonzero, so the zero-carrier cancellation branch has been ruled out, not omitted. Restore

$$f_{\text{old}} = f, \qquad p_{3,\text{old}} = (g-1)/2, \qquad n_{\text{old}} = n - b - g - 4f.$$

Then $Z_{\text{old}}/U + b = 2\tau$, the strong row is $f^2 - F_s = 0$, the auxiliary interface is unchanged, and $p_{3,\text{old}}(p_{3,\text{old}} + 1) = (G-1)/4$ restores the guard. Conversely, from every parent zero take $g = 2p_{3,\text{old}} + 1$ and

$$n_{\text{new}} = n_{\text{old}} + b + g + 4f_{\text{old}},$$

deleting f, p_3. The parent carrier supplies the integral τ as before; then one factor in (10) vanishes and the auxiliary row is unchanged. No additional Pell interpolation theorem is needed.

6.4 The exact shear and actual degree witnesses

Relative to each frozen Round18 equation, the change is simply

$$n_{\mathrm{old}} = n_{\mathrm{new}} - (2L + k).$$

Since L, k are independent of n, this is a polynomial triangular automorphism of the entire input lattice, with the inverse adding $2L+k$. It replaces the old carrier by $Z^2-4U^2D_t$ and its old centered square by Z^2. Direct expansion yields exactly the new norms and finalizers. This polynomial zero-set bijection is distinct from the root-dependent restoration to fifteen coordinates above.

Put

$$q_0 = 6\beta j, \quad X_0 = 12q_0 w, \quad Y_0 = v, \quad R_0 = q_0^4 z - 12q_0^3 wC, \quad U_0 = 8q_0^3 X_0 Y_0, \quad L_0 = X_0 Y_0^2 R_0.$$

The complete highest norm forms are

$$(N_2)_{60} = U_0^4 R_0^4 (q_0^8 + 4X_0^2 Y_0^4)^2,$$
$$(N_3)_{120} = 2^8 U_0^8 R_0^8 (q_0^8 + X_0^2 Y_0^4)^4.$$

The strong and auxiliary degrees in the fourteen-input chart are $12, 16$; the thirteen-input auxiliary degree is 36. The literal coefficient witnesses are

$$[j^{56} w^8 v^8 z^5 r^2 c^4 f^4 y] P_{14} = -2^{87} 3^{64} \beta^{56} \neq 0,$$
$$[j^{106} w^{14} v^{14} z^9 r^4 c^8 y] P_{13} = 2^{171} 3^{120} \beta^{106} \neq 0.$$

Hence the degrees are exactly $88, 156$ for the authentic fixed tuple, not merely generic-port bounds. The operation breakdowns are respectively $(64, 23, 23)$ and $(70, 23, 24)$ for multiplication, addition and subtraction. All unknowns and all six coefficient ports are live.

6.5 A scoped obstruction to degree reduction by this method

Let a further polynomial shear be $n \mapsto n + \Theta$, with Θ independent of n. If $\deg \Theta \leq 9$, let Θ_9 denote its degree-nine part. In the three-root chart the degree-60 inner norm is

$$U_0^4 \left[(\Theta_9^2 + 4R_0^2 q_0^8 - 4L_0^2)^2 + (8R_0 q_0^4 L_0)^2 \right].$$

This real-polynomial sum of squares is nonzero because its second square is nonzero. The outer subtracted term has degree at most 114, so it cannot cancel the degree-120 square. For the two-root chart, replace $4R_0^2 q_0^8$ by $R_0^2 q_0^8$ and $8R_0 q_0^4 L_0$ by $4R_0 q_0^4 L_0$; its degree-60 norm is likewise nonzero. If $m = \deg \Theta > 9$, the highest powers of $Z + U\Theta$ dominate and the norm degrees increase to $4(m+6)$ or $8(m+6)$. Thus no polynomial n-shear alone lowers 88 or 156. This explicitly restricted obstruction is not a lower bound over all encodings.

6.6 Verification and interpretation

The independent checker imports neither builder nor circuit code. It checks every named cut, topology, liveness and count; propagates complete leading forms; proves the shear, signed-factor, extraction and obstruction identities; and rejects each of the 227 separately modified live gates. Normal and optimized Python runs have the same acceptance conditions. All 615,883 raw bytes of the six fixed numerals are authenticated and independently reconstructed by the retained finite

recipe. The full inherited root/source packet is hash-pinned. The earlier carrier leading-form correction is already known and is not re-presented as a discovery.

Finite algebra does not decide existence of any zero. The unbounded square-class argument is the proof above; the signed-Pell/native/compiler and syntactic-refutation theorem is the explicit inherited ordinary mathematical dependency. No publication or new end-to-end proof-assistant or object-language consistency certificate is claimed.

7 Reproduction and the exact proof boundary

The archive supplies the literal residual lists and full expansions for each direct quartic, complete arithmetic DAGs for the small-variable families, all coefficient payloads and finite recipes, inverse maps, source hashes and ordinary proof dependencies. Its top-level replay works in fresh disposable copies in normal and optimized Python. Exact finite checks establish the actual polynomial identities, global leading forms, liveness, source bindings and deliberately corrupted negative controls. They do not replace any unbounded Pell or logical theorem by testing finitely many inputs.

The direct integer and natural branches use the numerical C222 calculus and its ZF/ZFC refutation calibration, not K215 simulation or Jones universality. The positive and root branches retain their different, explicit native coefficient calibrations. Whole tuples must be retained: matching names, bit sizes or elementary inequalities do not justify mixing a port from one source with another.

The repeated coordinate and residual operations have distinct meanings. An integral triangular shear is a bijection of an unrestricted integer lattice. Unit affine elimination gives a uniquely reconstructed coordinate. A change of the natural horizon or a canonical cofinal witness choice is *not* an arbitrary natural-coordinate bijection; it requires the new soundness and completeness arguments supplied in the natural section. Likewise, alternate coherent coefficient specializations are different integer polynomials, not identities with their predecessors.

The release includes independently performed mathematical and finite artifact reviews within this AI-assisted research workflow. These are not claims of outside human peer review or a new closed-kernel proof. No upstream repository was modified. No literature-wide priority, intrinsic arithmetic-complexity lower bound or globally simplest Diophantine equation is claimed. The limited row-span and coordinate-shear obstructions are stated only for the transformations they actually cover.

A A separately scoped whole radix47 specialization

All four lower-Y charts also admit an explicit specialization to the *whole incoming complemented-label radix47 tuple*. The sibling radix67 component remains frozen. This is not the current repository's conditional three-anchor radix47 layout, nor a newly claimed encoding or controller. The source is the incoming *Common Remainders–Radix47* package, whose complete ordinary proof, literal P66 equation, full coefficient payloads, 51 native target identities, independent review, and 37 retained source files are copied into this sidecar.

The four exact profiles remain $78/516, 79/382, 79/456, 80/338$, with ten strictly positive unknowns. Their free-port row lists and leading forms are unchanged; the actual specialized integer polynomials differ because all four fixed integers differ. The new tuple occupies 207,214 canonical raw bytes rather than 295,388, a reduction of 88,174 bytes; its largest port has 415,904 bits rather than 592,884. This coefficient improvement belongs to the incoming encoding. The present contribution is the explicit checked composition with the four new charts, not a new compression or

global-simplicity claim.

A.1 The exact incoming source and its changed native theorem

The inspected archive is pinned at snapshot `3ecd6a6cf53d47519228328567f45fb7ee9781d9`; its retained source snapshot is `55d1f72aedebf892e446c90d35fa1ae37689b0f2`. The full ordinary proof is `provenance/radix47/radix47/radix47_proof.md`. It complements all horizontal 8-bit and vertical 9-bit label codes, preserves all 943 indexed tiles and physical support, and gives all seven shared marker positions weight 2^{44}. It proves a predecoded cyclic carry cap 29, then a decoded cap 12. The occupied clause's outgoing carry 1 is retained, while separate exact predecessor resets protect phase anchors, copy tests, both clauses, padding, and cell origins. Its inverse color bijections recover the retained native language for arbitrary source words. These are unbounded ordinary arguments, not conclusions from a finite test set.

Let $\rho = 2^{47}$, $d = 47 \cdot 8807 = 413929$, and $B_{\text{base}} = 2^d$. The full new tuple, printed in the coefficient appendix, satisfies

$$\beta = B_{\text{base}} - 1, \quad \operatorname{pc}(M_C) = d - 986, \quad \operatorname{pc}(M_F - \beta) = 986,$$

$$M_C \equiv 2 \pmod 4, \quad M_F - \beta \equiv 4 \pmod 8, \quad 0 < M_C < \beta < M_F < 2\beta.$$

The source mask is shifted by β exactly once. Changing only the radix of the old formula would *not* give this tuple: selector coefficients, shared marker weights, and padding all change coherently.

The incoming fresh controller has prime $\lambda = 1012691$ and the following actual whole-tuple associations, all independently recomputed here:

Modulus	B_{base}	K_0	$1 + B_{\text{base}}K_0 + B_{\text{base}}^2$	Exact order of B_{base}
47	12	5	17	23
8807	2677	4079	5042	4403
1012691	959313	1005855	834350	1012690

Every bracket and $B_{\text{base}}^2 - 1$ is a unit at its listed prime. The lift

$$B_{\text{base}}^{1012690} \equiv 1 + 282338\lambda \pmod{\lambda^2}$$

has valuation exactly one. No coefficient association from radix55 or a different radix47 layout is imported.

The incoming cofinal width/height construction and safe-dummy subset sums tune the *actual* defining rank. From a refutation they provide genuine new native words with

$$q = B_{\text{base}}^N, \quad R > 3q, \quad R \equiv 3 \pmod 4, \quad \operatorname{pc}(R) = 3dN + 2,$$
$$X = 2^R = qW_{\text{native}}, \quad q^3 \mid Y, \quad Y/q^3 \text{ positive odd},$$
$$(K_0 + W_{\text{native}})C - F = t(q-1), \qquad t \geq 0.$$

Here Y is the required half-binomial and $t = z_{\text{source}} - 1$ for the constructed positive source transport coordinate. The incoming construction also gives strict margins $0 < F < q/\rho^{15}$, $0 < C < q/\rho^{5510}$ and $\alpha = q - F - C > 0$ under every safe dummy subset.

A.2 An explicit positive bridge, not a port-box substitution

The root, sign, first-index, early projection, and full-denominator arguments of the lower-Y reduction use only the scalar/phase/population interface and the retained universal Pell lemmas until the final native step. All those hypotheses hold for this *whole* tuple. Thus a zero yields positive outer data with $X = qW_0 = 2^R$, $q^3 \mid Y$, $R \equiv 3 \pmod 4$, the native half-binomial Y, and $t \geq 0$. Set $z_s = t + 1 > 0$; then

$$(K_0 + W_0)C + (q - F) - z_s(q - 1) = 1.$$

The defining rank is unchanged, since $h(q - F) + \alpha = q(q - F) - C$.

Reconstruct *fresh* source witnesses at

$$a_s = Y(X + 1), \quad A_s = a_s + 2, \quad H_s = 4a_s + 3, \quad \Delta_s = A_s^2 - 1 = a_s^2 + H_s.$$

The retained uniform source reconstruction supplies positive $c_s, k_s, \tau_s, \eta_s, \zeta_s, \gamma_s$ with

$$\begin{gathered} c_s = \psi_{A_s}(R) = k_s Y + \eta_s, \quad k_s = \eta_s + \zeta_s, \quad 0 < \eta_s < k_s, \\ \tau_s^2 - XY^2(XY^2 + 1)k_s^2 = 1, \quad h_s = (k_s - R - 1)/(XY) > 0, \\ D_s = X + a_s c_s + \gamma_s H_s = \chi_{A_s}(R). \end{gathered}$$

A fresh strong pair and the auxiliary-completeness theorem give

$$\begin{gathered} f_s^2 - \Delta_s i_s^2 c_s^4 = 1, \quad S_s = \Delta_s i_s c_s^2, \\ T_s = \frac{V_s + c_s + R f_s^2}{c_s f_s^2} > 0, \quad V_s = f_s^2(c_s T_s - R) - c_s, \\ S_s^2(V_s^2 - y_s^2) + y_s^2 = 1. \end{gathered}$$

All displayed source coordinates are positive integers. Feed the incoming P66 kernel its literal inputs

$$J, F, \alpha, z_s, f_s, h_s, i_s, T_s, Y/q^3, W_0, \tau_s, \eta_s, \zeta_s, y_s, \gamma_s.$$

With $\ell_s = (XY)(k_s Y)$, its *actual six factors* are

$$\begin{aligned} N_0 &= \tau_s^2 - \ell_s(\ell_s + k_s) = 1, \\ N_1 &= D_s^2 - \Delta_s c_s^2 = 1, \\ N_a &= S_s^2(V_s^2 - y_s^2) + y_s^2 = 1, \\ N_k &= k_s - h_s XY - R = 1, \\ N_t &= (K_0 + W_0)C + (q - F) - z_s(q - 1) = 1, \\ N_s &= \Delta_s f_s^2 - S_s^2 = \Delta_s. \end{aligned}$$

Therefore its literal polynomial $N_0 N_1 N_a N_k N_t N_s - \Delta_s$ vanishes with the entire incoming47 tuple. Its own ordinary P66 theorem, using the *new* carry/decoder proof, yields a finite syntactic refutation. We do not identify A_s with our auxiliary A, reuse a strong ordinate numerically, or invoke the old radix67 decoder. The fresh source witnesses are proof constructions, not extra unknowns of our ten-variable equation.

A.3 Constructive converse and exact finite closure

A refutation is fed to the incoming47 controller, not the old67 controller. Its $q = 2^m$ and actual $R > 3q$ imply $R > 3 \cdot 2^m \geq 6m$, hence $q^6 \mid X$. For the q^2 chart take $w = X/q$, $s = Y/q^2$; for the new q^1 chart take $w = X/q^2$, $s = Y/q$. These are positive integers, and in both cases $W_0 = W_{\mathrm{native}}$, preserving the coefficient in the actual transport. Now use the lower-Y reduction's fresh scale-one main/first Pell witnesses, positive seed period, doubled strong pair and unshifted positive T. The resulting integral ρ, κ, σ obey

$$n = gy + (1 + hL)t + \rho - \kappa - \sigma > 0,$$

because the seed is chosen with $g > \kappa + \sigma$. Its carrier is gy and the new polynomial vanishes. This constructs every independent input positively; it is not a pointwise packed-word identification or inverse shear.

The sidecar rebuilder imports no inherited producer. It decodes the entire seed, reconstructs all 943 tiles and 3297 coefficients, checks all four payloads against the whole incoming P66 binding, collects all 51 target linear identities, and checks the actual mask populations, carry inequalities, primes, exact orders, units and lift. Its graph checker proves all leading forms, all row identities, liveness and counts and rejects actual algebraic corruptions. Normal and optimized receipts agree. The complete ordinary bridge is in `PROOF.md`; `REDUCTION_PROOF.md` is the unchanged mathematical reduction with its original67 context explicitly overridden only by this bridge.

The conclusion remains the ordinary ZFC-provable equivalence between no positive zero and Con(ZFC) at the retained calibration boundary. No new full Lean/Rocq or object-ZFC derivation, consistency decision, repository acceptance of the incoming47 encoding, or global optimum is asserted.

B Complete complemented-label radix47 constants

This is the entire incoming complemented-label tuple, not the repository's conditional three-anchor radix47 layout. All indices, labels and sums below are fixed data. The finite recipe and 436-character seed are reproduced from `provenance/radix47/radix47/equation.tex`; its K is renamed K_0. The independent reconstruction checks all 943 tiles, all 3297 coefficients, and every byte of all four actual fixed integers. No new encoding is claimed. **The four fixed integers.** Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{47}, \quad B = \rho^{8807}, \quad \beta = B - 1,$$
$$G_m(x) = \sum_{j=0}^{m-1} x^j, \quad \Lambda = \rho^{4393} + \rho^{5495}.$$

$$\begin{aligned}
M_C &= \beta + \rho - (\rho + 3)\rho^{1063} \\
&\quad - \sum_{e=0}^{3296} \lfloor (D_\rho(\Phi(\rho), e) + \rho - 1)/\rho \rfloor \, \rho^e, \\
M_F &= 2B + 3 + (2^{47} - 2)\Lambda \\
&\quad + \rho^{1013} G_{10}(\rho) + \rho^{1065} G_9(\rho) + \rho^{2195} + \rho^{3296} \\
&\quad - \rho^{8792} + (2^{28} - 3)\rho^{7694} - \rho^{7691}, \\
K_0 &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\
&\quad + \Lambda \Phi(1/\rho),
\end{aligned}$$

where

$$\begin{aligned}\Phi(x) = \sum_{i=0}^{942} &(1 + 2^{10} + 2^{10}w_i + 2^{18}n_i \\ &+ 2^{27}p_i + 2^{35}s_i)x^i \\ &+ 2^{10}x^{1003}(x^{20} + 2^{17})G_8(2x) \\ &+ 2^{18}x^{983}(x^{30} + 2^{17})G_9(2x) \\ &+ 2^{44}x^{992}(1 + x^{19} + x^{30} + x^{39}) \\ &+ 2^{44}x^{1095}(1 + x^{1100} + x^{2201}).\end{aligned}$$

The rational intermediate in K_0 cancels exactly; all four results are positive integers.

The 943 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. For $d = 0, 1$, let

$$h(q, b) = 2q + b + 2, \quad \mathcal{S}_d = \{u_j : d_j = d,\ 0 \le j < 430\},$$
$$r_{d,q} = 205 - 199d + |\mathcal{S}_d \cap \{0, \dots, q-1\}|.$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 942$; all index ranges are fixed:

index i	tile $t_i = (w_i, n_i, p_i, s_i)$
$0, 1$	$(2, 435, 3, 1),\ (5, 0, 0, 0)$
$j + 2\ (j = 2q + b < 430)$	$((1 - d_j)r_{0,u_j}, h(q, b), d_j r_{1,u_j}, v_j)$
$434 - 2a + b\ (a, b = 0, 1)$	$(0, 432a + b, 0, 432a + b)$
$\substack{424+2r_{d,q}+b \\ q\in\mathcal{S}_d,\ d,b=0,1}$	$(dr_{d,q}, b, (1 - d)r_{d,q}, h(q, b))$
$924 + 2a + b\ (a, b = 0, 1)$	$(0, 432a + b, 0, 438)$
928	$(2, 434, 2, 0)$
929	$(3, 436, 4, 4)$
930	$(4, 436, 4, 0)$
$497 + t\ (t = 434, 435, 436)$	$(1, 438, 1, t)$
$934 + 3a + b\ (a, b = 0, 1, 2)$	$(a^2, 437, b, 437)$

A closed decoder for the 430 actions. Put $D_b(A, k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \left\lfloor \mathcal{A}/2^{636} \right\rfloor \bmod 2^{1675},$$
$$E_0 = \left\lfloor \mathcal{A}/2^{2311} \right\rfloor,$$
$$\nu_j = D_2(N, j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430 + q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14 + m_q(D_{16}(E_0, \tau_q) - 14),$$
$$u_j = \nu_j(1 + \sigma_j) + (1 - \nu_j)D_{216}(T_0, j - \sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0, \ldots, 61$, is

`0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz`.

Concatenate the following 436 characters as digits $e_0, \ldots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```